

University of Information Technology and Sciences
Faculty of Science and Engineering
Department of CSE
Mid Term Examination, Spring- 2024
Course Title: Coordinate Geometry and Vector Analysis.
Course Code: MAT261

Full Marks: 20

Time: 1 hour

Answer all questions.

1. (a) The coordinates of A, B, C, D are (6,3), (-3,5), (4,-2) and (x,3x) respectively and $\frac{\Delta DBC}{\Delta ABC} = \frac{1}{2}$, Δ is the area of the triangle, Find x. [2]
- (b) (i) Transform to Cartesian co- ordinates the equation $r(\cos 3\theta + \sin 3\theta) = 5k \sin \theta \cos \theta$. [3]
(ii) Change the equation to polar co-ordinates $x^4 + x^2 y^2 - (x + y)^2 = 0$.
- (c) Reduce the equation $32x^2 + 52xy - 7y^2 - 64x - 52y + 148 = 0$ to standard form. [5]
confirm type of conic.
2. (a) Find the transformed equation of $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$, when the axes are rotated through an angle $\frac{\pi}{6}$. [3]
- (b) Determine the equation of the curve $x^2 + 2y^2 - 6x + 7 = 0$ when origin is transferred to the point (3,1) [3]
- (c) Prove that the equation $2y^2 - xy - x^2 + 2x + y - 1 = 0$ represents a pair of straight lines, find also their point of intersection and angle between them. [4]

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Term Final Examination, Spring-2024
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Marks: 50

Time: 3 Hours

(Answer all the questions)

1. (a) Test the nature of the conics given by the following equations: [04]
i) $x^2 - 6xy - 2y^2 - 4y + 6 = 0$
ii) $4x^2 + 9y^2 + 36y - 8x - 31 = 0$
(b) Reduce the equation $5x^2 - 24xy - 5y^2 + 58y + 4x - 59 = 0$ to its standard form. [06]
2. (a) Define direction cosines and direction ratios of a line. Find the direction cosines l, m, n of two lines which are connected by the relations, [05]
 $l - 5m + 3n = 0, 7l^2 + 5m^2 - 3n^2 = 0.$
(b) What is plane? Find the equation of the plane passing through the points $(1, 0, -1)$, [05]
 $(2, 1, 3)$ and is perpendicular to the plane $2x + y + z - 1 = 0.$
3. (a) The following forces act on a particle P. Find the magnitude of the resultant vectors [05]
 $\vec{F}_1 = 2\hat{i} + 3\hat{j} - 5\hat{k}, \vec{F}_2 = -5\hat{i} + \hat{j} + 3\hat{k}, \vec{F}_3 = \hat{i} - 2\hat{j} + 4\hat{k} \text{ \& } \vec{F}_4 = 4\hat{i} - 3\hat{j} - 2\hat{k}.$
(b) What is dot and cross product? Find the directional derivatives of $\phi = 4xz^3 - 3x^2y^2z$ [05]
at $(2, -1, 2)$ in the direction $2\hat{i} - 3\hat{j} + 6\hat{k}.$
4. (a) Define gradient and Curl. Find $\nabla\phi$ if $\phi = \ln|\mathbf{r}|$ and also prove that $\vec{\omega} = \frac{1}{2} \text{curl } \vec{v}.$ [06]
(b) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the [04]
time. Find the components of its velocity and acceleration at time $t=1$ in the direction $\hat{i} - 3\hat{j} + 2\hat{k}.$
5. (a) If $\phi = 2xyz^2, \underline{F} = xy\underline{i} - z\underline{j} + x^2\underline{k}$ and C is the curve $x = t^2, y = 2t, z = t^3$ from [06]
 $t = 0$ to $t = 1$, evaluate the line integrals
(i) $\int \phi d\underline{r}$ (ii) $\int \underline{F} \times d\underline{r}$
(b) State Green's Theorem. Verify the Green's theorem in the plane for [04]
 $\oint_C (xy + y^2) dx + x^2 dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2.$